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1 AUGUST 1962
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16 DECEMBER 1957

MILITARY SPECIFICATION

INDICATOR, ATTITUDE, SPHERICAL, DIRECT READING, FAST ERECTING, ELECTRIC GYRO

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers certain design and all performance requirements for procurement of spherical, direct reading, fast erecting, electric gyro attitude indicators in 5 by 5-1/4 inch cases.

* 1.2 Classification - The Attitude indicators shall be of the following types:

Type I - Indicators without integral lighting and without synchro output. (MS28086-1)

Type II - Indicators without integral lighting and with synchro output. (MS28086-2)

Type III - Indicators with integral lighting and without synchro output. (MS28086-3)

Type IV - Indicators with integral lighting and with synchro output. (MS28086-4)

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of the invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

DD-G-451	Glass, Flat and Corrugated, for Glazing, Mirrors and other uses
QQ-C-320	Chrominum Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-C-5015	Connectors Electric, "AN" Type

FSC 6610

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MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-N-6011	Nitrogen, Liquid and Gas
MIL-L-6085	Lubricating Oil, Aircraft Instrument, Low Volatility
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-M-7793	Meter, Time Totalizing
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-R-22973	Reliability Index Determination for Avionic Equipment Models, General Specification for
MIL-L-25467	Lighting, Integral, Instrument, General Specification for
MIL-G-25591	Gyroscope, Rate, Switching Type MC-1
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS28086	Indicator, Attitude, Spherical, Direct Reading, Fast Erecting, Electric Gyro
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33586	Metals, Definition of Dissimilar

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2.2 Availability of Documents -

(1) When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

* 3.1 Qualification - The attitude indicator furnished under this specification shall be a product which has been tested, and has passed the qualification tests specified herein and has been listed on, or approved for listing on, the applicable qualified products list.

3.2 Selection of Government Documents - Specifications and Standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial Parts - Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are not suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties, may be used provided:

(a) The can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and if practicable, on the contractor's drawings.

3.2.2 Standard Parts - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended.

3.3.1 Critical Materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Metals - Metals shall be of the corrosion resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray or atmospheric conditions to which the instrument may be subject when in storage or during normal service life.

3.3.2.1 Dissimilar Metals - Dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings or hermetic sealing.

3.3.3 Non-magnetic Materials - Non-magnetic materials shall be used for all parts of the instrument except where magnetic materials are essential.

3.3.4 Fungus-Proof Materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used in a hermetically sealed case, fungicidal treatment will not be necessary.

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3.3.5 Corrosive Fumes - Materials used in the interior construction of the instrument shall not produce corrosive or deleterious fumes.

3.4 Design and Construction - The indicator shall be designed to indicate the attitude of an airplane throughout all maneuvers and to permit loops and rolls to be made with controlled tumbling of the gyro. The instrument shall be so constructed that no parts will work loose in service and will withstand the normal strains of jars, vibrations and such other conditions as are incident to shipping, storage, installation and service without failure.

3.4.1 Maintenance - The design shall be reasonably simple to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly by those tools and items of maintenance equipment which are normally available as commercial standards without requiring special tools and fixtures.

* 3.4.2 Case - The outline dimensions of the indicator case shall conform to MS28086. The case shall be suitable for front mounting on an instrument panel. The body and mounting flange shall be made of non-ferrous, low density metal, uniform in texture and have a smooth surface. The body and flange shall be finished with a lusterless, black material, Color No. 37038 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service.

* 3.4.2.1 Hermetic Sealing - The case shall provide a hermetically sealed enclosure for all of the mechanism. The case shall be so constructed that it may be opened, the mechanism removed and replaced and the case reclosed and resealed at least three times. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any atmosphere to which the instrument may be subjected in normal use in military aircraft.

3.4.2.1.1 Leak Rate - The indicator shall be so sealed that the maximum leak rate will not permit more than 5 percent contamination of the filling medium after 3000 hours at a pressure differential equivalent to 100,000 feet altitude.

3.4.2.2 Filling Medium - The filling medium shall be a mixture of 90 percent nitrogen and 10 percent helium or a filling medium approved by the procuring activity. The nitrogen used shall be of a quality in accordance with Specification MIL-N-6011, Grade A, Type I. The filling medium shall be of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The case shall be filled to an absolute pressure of 1 ± 0.1 atmosphere.

* 3.4.3 Pitch Trim Adjustment - A pitch trim knob shall be mounted on the front lower right hand side of the indicator. The knob shall have approximately 3/4 turn of total freedom. The freedom in each direction shall be limited by a positive stop. When the knob is rotated to the stop in the clockwise direction, the miniature airplane shall move up a minimum of 11 degrees. When the knob is rotated to the stop in the counterclockwise direction, the miniature airplane shall move down a minimum of 15 degrees. The pitch trim mechanism shall have sufficient friction to remain fixed when set but it shall move smoothly without evidence of binding, resiliency or erratic motion.

3.4.4 Gyro Erection

3.4.4.1 Initial Erection Device - The instrument shall incorporate a fast erection system to bring the gyro to the gravity vertical when power is initially applied.

3.4.4.2 Normal Erection - The instrument shall incorporate an erection system to maintain the gyro in its gravity vertical position during normal operation.

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3.4.4.3 Demand Erection Device - A pushbutton shall be provided on the face of the instrument to actuate a fast erection system when the instrument is operating. Momentary depression of the pushbutton shall start the fast erection and bring the gyro to its gravity vertical from any random position in pitch and bank. The pushbutton shall be marked "Push For Fast Erection".

3.4.4.4 Test Connections - Connections to the roll and pitch liquid levels, the roll and pitch torquer control fields, and the cutoff turn sensitive switch shall be brought out to the AN connector at the rear of the instrument so that a test box properly connected may be used to cut off the erection, allow the instrument to erect normally, or process the gyro to any desired position. During normal operation jumpers will be used to complete the erection circuits brought out to the AN connector.

* 3.4.4.5 Erection During Turns - An erection cutoff system controlled by a turn sensitive switch shall be employed for minimizing turn error in the indicator. When turns are made at rates sufficient to displace the conducting medium in the turn sensitive switch in excess of 6 degrees, bank erection control shall be cut off. Provision shall be made for the incorporation of an external rate gyro in accordance with Specification MIL-G-22591, Type MC-1, for control of erection during turns, without internal wiring change to the instrument.

* 3.4.5 Weight - The indicator shall be so designed that when completely assembled the weight shall not exceed 8.5 pounds.

3.5 Electrical -

3.5.1 Electronic Parts - Electronic parts and the application thereof shall be in accordance with Specification MIL-E-5400. Parts that do not appear on approved lists in accordance with Specification MIL-E-5400 shall not be used unless approved by the Bureau of Naval Weapons.

3.5.2 Wiring - Internal wiring shall be color coded, neat and accomplished in such a manner that individual wires may be easily traced. Wires shall be secured mechanically to the terminals prior to soldering, where practicable. The wiring shall be insulated from the instrument case except that Pin G may be grounded to the case.

* 3.5.3 Electrical Receptacle - The electrical receptacles used shall conform to Specification MIL-C-5015 and shall be of types MS3102-20-29P and MS3102-18-1P. The electrical receptacle code markings specified on MS28086 shall be legibly and permanently applied to the indicator case.

* 3.5.4 Power Requirement - The indicator shall be designed to operate within the range of 180 to 220 volts and 380 to 420 cycles in accordance with the characteristics in MIL-STD-704.

3.5.4.1 Phase - The phase rotation shall be ABC with phase A to pin A, phase B to pin B, phase C to pin C, and neutral to pin N. The power input connections shall conform to Tables I and II.

3.5.5 Synchro Excitation - The pitch and roll synchro rotor leads shall be isolated from the case or power leads and shall be made available at the AN connector. When the rotors are excited with phase C on pins H and J and neutral on pins G and I, the outputs and sensing shall be in accordance with paragraph headed "Pitch and Roll Outputs".

3.5.6 Pitch and Roll Outputs - The electrical signals in pitch and roll shall be synchro type with nominal maximum of 11.8 volts line to line output. The outputs shall be continuous in 360 degree roll and plus or minus 82 degree pitch maneuver. Each synchro, pitch and roll, shall be capable of operating up to five Eclipse Pioneer AY 500-5 (or approved equivalent) type control transformers each loaded with a 10,000 ohm load. The outputs shall be set at electrical zero with a null between pins A and B in roll and D and E in pitch with the gyro level. The output signal from pins A to B shall be in-phase with phase C to neutral when the control is in a left bank maneuver. The output signal from pins D and E shall be in-phase with phase C to neutral when the control is in a climb maneuver.

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TABLE I

20-29 P Electrical Pins	Connections for Types I, II, III and IV
A; B; C N	115/200 volt, 400 cps power source (neutral - N)
L; G	5 volt integral lighting source with pin G grounded to the case. Applies to Types III and IV.
E	This pin is to be used for supplying power to the erection torque motors. During normal operation (cutoff by turn sensitive switch) this pin shall be connected (externally) to pin B. For cutoff by rate gyro, the rate gyro may be used to open the connections to this pin thereby cutting the erection.
S; R	These two pins are to be connected across the turn sensitive erection cutout switch and should be shorted externally by a jumper during erection cutout by rate gyro.
	The following sets of pins are to be used for precessing the gyro during laboratory tests. During normal operation these pins should be shorted as follows:
D; F H; J	Roll channel (D shorted to F, and H shorted to J)
K; M P; T	Pitch channel (K shorted to M, and P shorted to T)

TABLE II

18-1 P Electrical Pins	Connections for Type II and Type IV
A; B; C	Roll axis transmitting synchro stator
D; E; F	Pitch axis transmitting synchro stator
G: H	Synchro exitation, roll (26 volts ac, 400 cps.)
I; J	Synchro exitation, pitch (26 volts ac, 400 cps.)

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** 3.5.7 Time Totalizing Meter - The case shall incorporate a time totalizing meter conforming to the applicable requirements of Specification MIL-M-7793.

3.5.8 Power Warning Flag - A warning device shall be incorporated which will immediately indicate when phase rotation is incorrect, ac power failure occurs, or during initial and demand erection conditions.

3.6 Dial -

* 3.6.1 Cover Glass - The quality of the cover glass shall be in accordance with Specification DD-G-451, Type II, Quality AA. Any flaws permitted by Specification DD-G-451 shall not interfere with reading of the dial. The glass shall not be fluorescent when exposed to ultraviolet light or rays in the 365 millimicrons region. The diameter of the cover glass shall be $4.451^{+0}_{-0.005}$ inches and the thickness 0.140 ± 0.005 inch. The cover glass shall be properly sealed. There shall be no fogging due to the accumulation of a film of water, oil or any other substance on the inside surface of the cover glass under operational conditions.

* 3.6.2 Dial Markings - The dial shall be marked as shown in Figures 1 and 3. The style and proportions of the letters and numerals placed on the dial shall conform to MS33558. Unless otherwise specified, all colors shall be in accordance with FED-STD-595, with the paint being such that it diffuses light sufficiently to avoid "highlights" when illuminated. The background of the dial area shall be finished with lusterless black material of Color No. 37038 of FED-STD-595. The painting material used shall be of a quality sufficient to ensure durability of the markings in service use.

3.6.2.1 Visibility of Dial - The pitch scale graduations from 40 to -40 degrees, the bank index, the numerals, at least 1/16 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustrum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the instrument case.

3.6.3 Illumination of Dial - An integral lighting system that shall satisfactorily illuminate all dial markings uniformly but without glare shall be provided for Types II and IV indicators. The lighting system shall be in accordance with the requirements of Specification MIL-L-25467, except that the colors shall be as specified in Figure 3. The sources of illumination shall not be visible from any point within the frustrum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the instrument case.

3.6.4 Display - The display shall consist of an adjustable miniature airplane, a bank angle scale, a bank index, a moving two-colored spheroidal background with a pitch scale and a distinct horizon line dividing the two colors.

3.6.4.1 Miniature Airplane - The miniature airplane shall be in accordance with Figure 1.

3.6.4.2 Bank Angle Scale - The bank angle scale shall be in accordance with Figure 1. The scale shall be attached to the instrument case. The 5 and 15 degree markings shall be engraved and painted black so they will be visible for calibration purposes. As an alternate, calibration marks may be white Color No. 37778 of FED-STD-595 and 0.005 inch long.

3.6.4.3 Bank Index - The bank index shall be in accordance with Figure 1 and shall indicate the bank attitude within a range of from zero to 180 degrees in left and right bank. The member which moves the bank index shall be attached to the roll gimbal. The bank index shall not move with pitch deviations. The front surface of the index shall be in the same plane as the front surface of the bank angle scale. The small end of the bank index shall be within 3/64 of the zero mark on the bank angle scale when the two are adjacent.

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3.6.4.4 Spheroid - The spheroid shall conform to Figure 2 and shall be mounted in the roll gimbal of the indicator. It shall rotate with the gimbal about the roll axis for roll indications and it shall rotate in relation to the roll gimbal about the pitch axis for pitch indications. The roll axis shall be perpendicular to the face of the indicator. The pitch axis shall be horizontal when the indicator is in a zero roll attitude and the gyro is erect. The horizon line on the spheroid shall be established at the zero pitch position. The spheroid shall be driven in pitch in such a way that for a dive maneuver the horizon line goes above the miniature airplane, and for a climb maneuver the horizon line goes below the miniature airplane, when the miniature airplane is set to the horizon line during level flight. The spheroid shall indicate the pitch attitude within a range of from zero to 82 degrees in climb and dive. The surface of the spheroid shall move 0.035 (nominal) inch for each degree of pitch change. The colors and markings of the spheroid shall be in accordance with Figure 3.

* 3.7 Performance - The attitude indicator shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

3.8 Physical Requirements -

* 3.8.1 Protective Treatment - When materials are used in the construction of an instrument that are subject to corrosion in atmospheric or environmental conditions likely to occur during normal service life, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip or scale during normal service life or are affected by extremes of atmospheric or environmental conditions shall not be used.

3.8.2 Alluminum Alloy Parts - Where practicable, Alluminum Alloy Parts shall be covered with an anodic film conforming to Specification MIL-A-8625 except that the dial, if of alluminum alloy, small holes, pipe threads and case inserts need not be anodized. Alluminum alloys which do not anodize satisfactorily shall be coated with chemical film in accordance with Specification MIL-C-5541.

3.8.3 Iron and Steel Parts - Where practicable, iron and steel parts shall be nickel, chromium, or cadmium plated in accordance with Specifications QQ-N-290, QQ-C-320 and QQ-P-416, respectively. Parts in a confined space in the presence of organic material shall be tin-cadmium plated.

3.8.4 Screw Threads - Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL-S-7742.

3.8.5 Soldering - Soldering shall be performed in accordance with Specification MIL-S-6872.

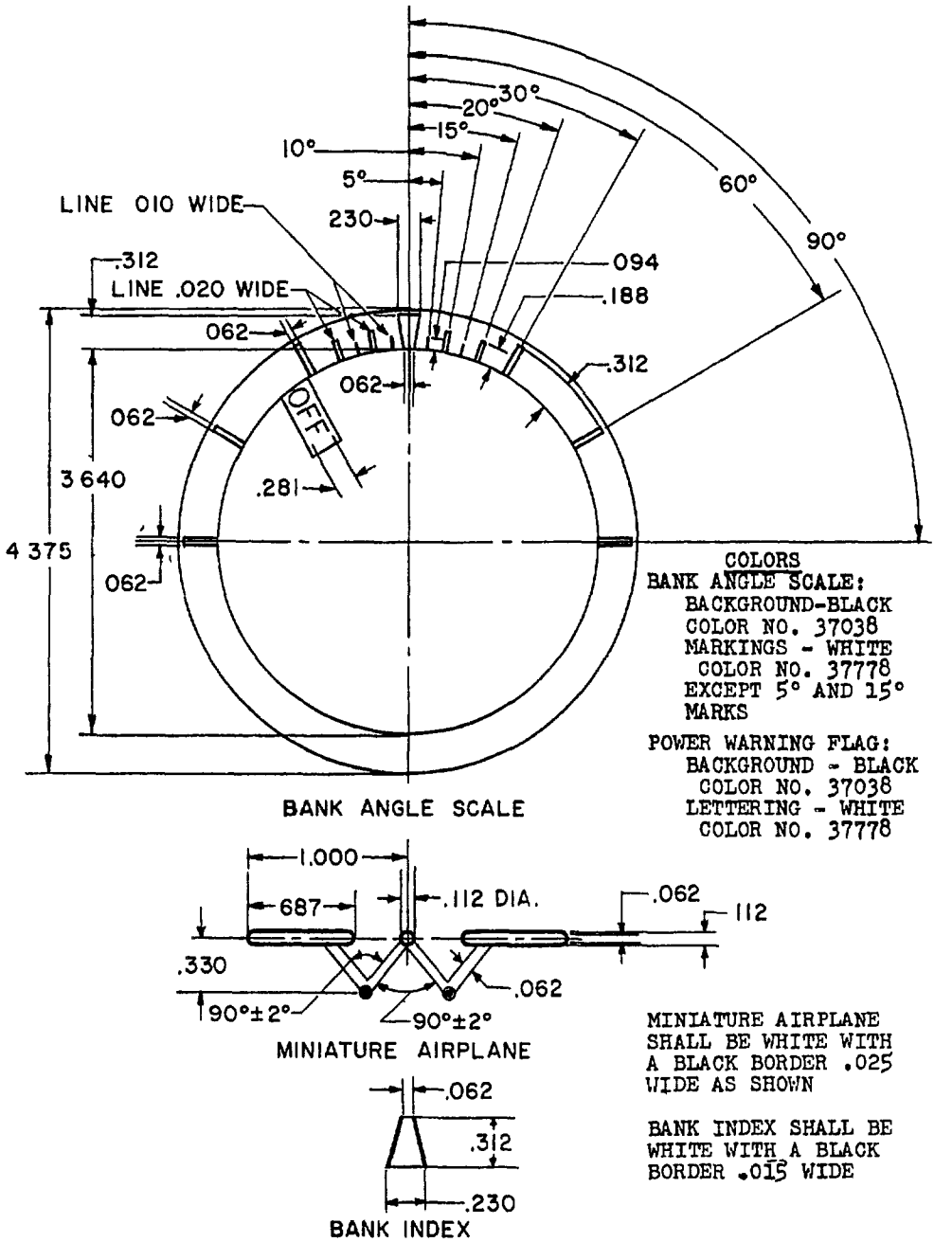
* 3.8.6 Lubrication The lubricating oil used shall be in accordance with Specification MIL-L-6085 and shall be sufficient for at least 1000 hours of operation.

* 3.8.7 Case Markings - All markings on the indicator case shall be made with material of Color No. 37778 of FED-STD-595. The style and proportions of the letters and numerals shall conform to MS33558. The marking material shall be of a quality sufficient to ensure durability of the markings in service use.

* 3.8.8 Dimensions - Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation or performance of the instrument, they shall be held or limited accordingly.

* 3.8.9 Gears - Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

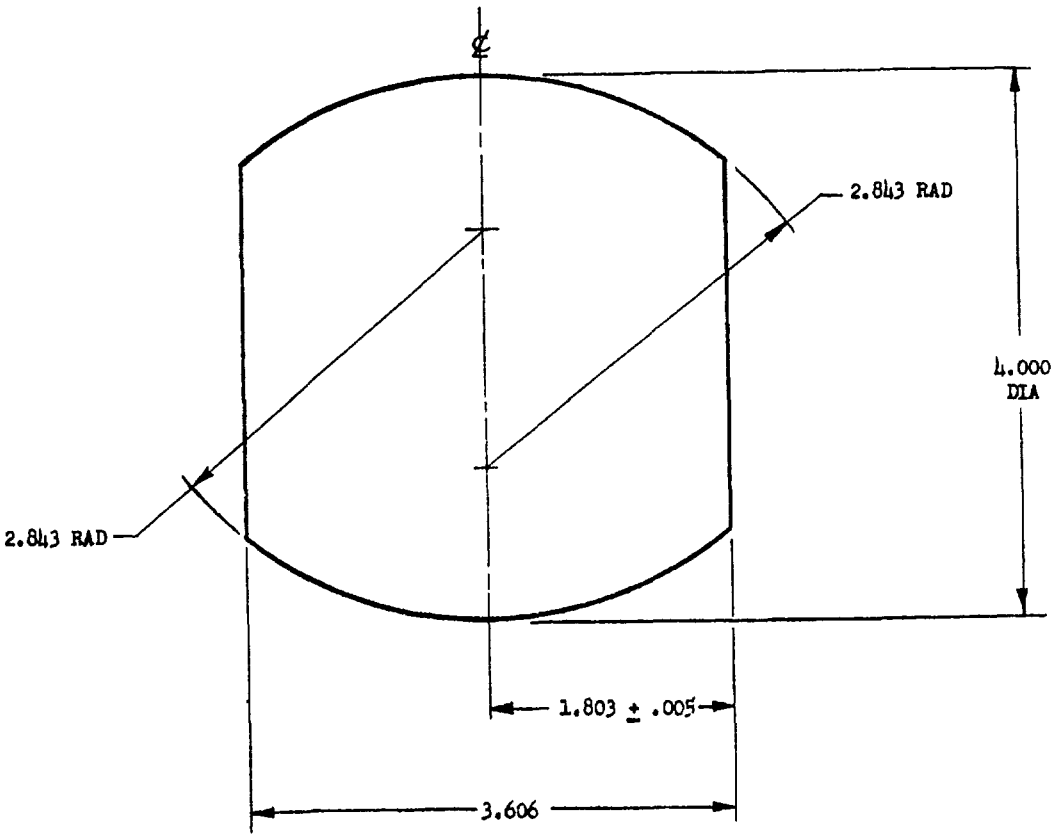
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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ± .005, ANGLES ± 1/2°.

FIGURE 1. BANK ANGLE SCALE, MINIATURE AIRPLANE AND BANK INDEX

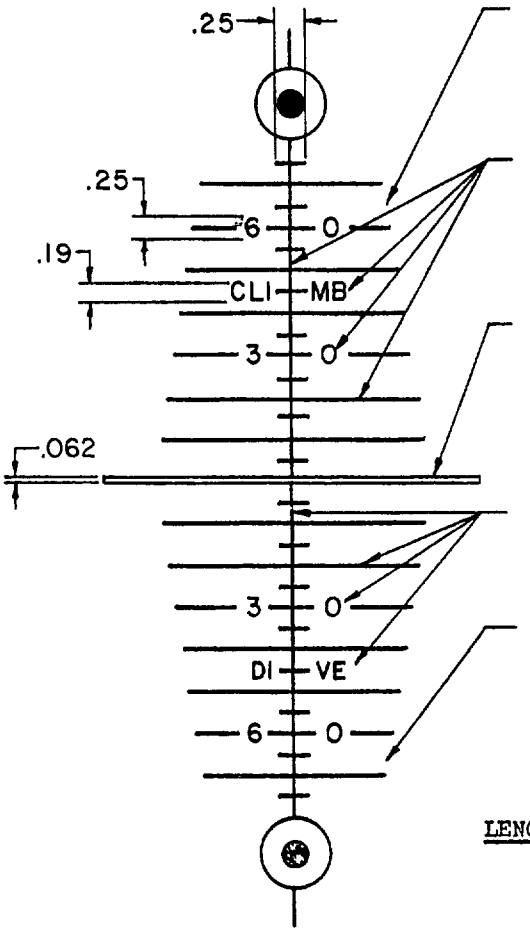
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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES ± .010.

FIGURE 2. SPHEROID, FRONT VIEW

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THE BACKGROUND OF THE SPHEROID ABOVE THE HORIZON LINE SHALL BE LIGHT GRAY, COLOR NO. 36373

ALL MARKINGS, NUMERALS AND LETTERS ABOVE THE HORIZON LINE SHALL BE BLACK, COLOR NO. 37038

HORIZON LINE SHALL BE WHITE COLOR NO. 37778, AND SHALL EXTEND COMPLETELY ACROSS THE DIAMETER OF THE SPHEROID.

ALL MARKINGS, NUMERALS AND LETTERS BELOW THE HORIZON LINE SHALL BE LIGHT GRAY, COLOR NO. 36373

THE BACKGROUND OF THE SPHEROID BELOW THE HORIZON LINE SHALL BE BLACK, COLOR NO. 37038

ALL MARKINGS, NUMERALS AND LETTERS SHALL BE .020 WIDE UNLESS OTHERWISE SPECIFIED

LENGTH OF MAJOR PITCH SCALE LINES

10° - - - -	2.50
20° - - - -	2.38
30° - - - -	2.25
40° - - - -	2.12
50° - - - -	2.00
60° - - - -	1.88
70° - - - -	1.75
80° - - - -	CIRCLE
85° - - - -	SOLID DOT
110° - - - -	END OF VERTICAL LINE

THE PITCH REFERENCE LINES SHALL BE PLACED ON THE SPHEROID SO THAT THEY WILL APPEAR STRAIGHT WHEN DIRECTLY BEHIND THE MINIATURE AIRPLANE.

THE CENTER OF THE SOLID DOTS SHALL BE THE ±90° PITCH POSITION AND FROM CENTER TO CENTER SHALL REPRESENT 180° OF PITCH INDICATION. THE PITCH INDICATION SHALL BE LINEAR OVER THE FULL RANGE AND THIS 180° RANGE SHALL BE SPACED OVER 180° OF THE SPHEROID SURFACE.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: TWO PLACE DECIMALS ± .02, THREE PLACE DECIMALS ± .005.

FIGURE 3. SPHEROID MARKINGS

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3.8.10 Workmanship - The instrument, including all parts and accessories, shall be so constructed and finished that it will be free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, plating, riveting, machine screw assemblies and freedom of parts from burrs and sharp edges.

* 3.8.11 Cleaning - The instrument shall be thoroughly cleaned of loose, spattered or excess solder, resin flash that may crumble, metal chips and other foreign matter, after final assembly.

3.9 Identification and Installation -

3.9.1 Nameplate - The nameplate shall be securely attached to the indicator case and shall be legibly and permanently marked with the following information:

BUWEPS - U. S. Navy
Indicator, Attitude, Spherical, Direct Reading,
Fast Erecting, Electrical Gyro, Type _____
MS28086- (Add proper dash number)
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract No.
Manufacturer's Name or Trademark

3.9.2 Manufacturer's Part Number - The manufacturer's part number marked in the space provided in the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

* 3.9.3 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL-D-70327.

* 3.9.4 Use of Military Designations - Military designations shall not be applied to a product, except for Qualification Test Samples, nor referred to in correspondence or sales matter, until notification has been received from the activity responsible for qualification.

* 3.9.5 Installation Instructions - The contractor shall furnish with each instrument one copy of the installation instructions with illustrations and diagrams. Prior to printing, two copies shall be furnished to the Bureau of Naval Weapons for approval. Whenever possible the instructions shall be arranged to require only one sheet of paper either 8-1/2 by 11 or 11 by 17 inches.

3.9.6 Mounting Screws - The contractor shall furnish sufficient mounting screws for installing the indicator. The screws shall be No. 10-32 NF-2 fillister head, nonmagnetic, machine screws having a durable and dull black oxidized, or dull black nickel finish. The length shall be sufficient for mounting on a panel 5/16 inch thick.

3.9.7 Envelope - An envelope furnished by the contractor containing installation instructions and mounting screws shall be packaged with each indicator, and shall be marked with the following information:

"IMPORTANT
THIS ENVELOPE CONTAINS INSTRUCTIONS AND MOUNTING SCREWS"

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Contractor's Responsibility - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Tests - The Inspection and Testing of the attitude indicator shall be classified as follows:

(a) Qualification Tests - Qualification tests are those tests performed on sample instruments submitted for qualification as a satisfactory product.

(b) Acceptance Tests - Acceptance tests are those tests performed on instruments manufactured and submitted for acceptance under contract.

* 4.3 Qualification Tests - The qualification tests of the instrument shall consist of all of the tests of this specification specified under the paragraph headed Test Methods.

* 4.3.1 Qualification Test Instructions - Qualification test samples shall consist of three attitude indicators, of each manufacturer's part number to be furnished in accordance with this specification. The instruments submitted for testing shall not have been previously tested except for the Individual Tests. The samples shall be forwarded to the testing laboratory designated in the letter of authorization.

* 4.3.1.1 Test Sample Identification - The test samples shall be plainly identified by securely attached, durable tags marked with the information listed below:

Sample for Qualification Testing
Indicator: Attitude, Spherical, Direct Reading,
Fast Erecting, Electric Gyro
Submitted by (Manufacturer's name, date) for
Qualification Testing in accordance with
Specification MIL-I-21203 under authorization (reference letter authorizing tests).
MS28086- (Add proper dash number)
Type Number _____
Manufacturer's Part Number

* 4.4 Acceptance Tests - The acceptance tests of the instruments shall consist of the Individual Tests and the Sampling Plans of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

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4.4.1 Individual Tests - The individual tests shall consist of those listed below, conducted in the order listed, on each instrument.

Inspection Test	Erection Rate Test
Initial Erection Test	Erection Cutoff System Test
Final Erection Test	Demand Erection Test
Power Warning Flag Test	Roll, Pitch, and Yaw Test
Low Voltage Operation Test	Drift Test
Power Interruption Test	Loop and Roll Test
Current Test	Dielectric Strength Test

* 4.4.2 Sampling Plans and Tests -

4.4.2.1 Sampling Plan A - Samples shall consist of three attitude indicators selected at random by the Inspector from each lot of 100 or less which have passed the Individual Tests. A lot shall consist of identical instruments with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at substantially the same time. Instruments which have been subjected to the Sampling Plan A tests shall not be delivered on the contract until they have been reconditioned and resubmitted to all Individual Tests.

4.4.2.1.1 Sampling Plan A Tests - The sampling plan A tests shall consist of those listed below, conducted in the order listed, on each sample.

Pitch Scale Sensitivity Test	Vibration Error Test
Synchro Transmitting Error Test	Low Temperature Operation Test
Illumination Test	High Temperature Operation Test
Magnetic Effect Test	Seal Test

4.4.2.2 Sampling Plan B - Unless otherwise specified, the samples shall consist of three attitude indicators selected at random by the Inspector from the first 15 instruments produced on contract. The samples shall be selected within ten days after manufacture and forwarded to the laboratory specified herein (see 6.2). Samples which have been subjected to the Sampling Plan B tests shall not be delivered on contract until they have been reconditioned and resubmitted to all the Individual Tests. Each sample shall be plainly identified by securely attached, durable tags marked with the following information:

Indicator, Attitude, Spherical, Direct Reading,
Fast Erecting, Electric Gyro, Type _____.
Submitted by (manufacturer's name, date) for production
acceptance Sampling Plan B testing in accordance with
Contract/Order No. _____
Manufacturer's Part Number

4.4.2.2.1 Sampling Plan B Tests - The sampling plan B tests shall consist of those listed below, conducted in the order listed, on each sample.

Sampling Plan A Tests

Radio Noise Interference Test

Low Temperature Exposure - High Altitude Tests

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~~High Temperature Exposure Test~~

Vibration Failure Test

4.4.3 Reliability -

* 4.4.3.1 Mean Time Between Failures (MTBF) Determination - Tests shall be performed to determine the MTBF of the indicator. Specification MIL-R-22973 shall be used as a guide for the determination. The procedure to be followed is outlined in the following paragraphs.

* 4.4.3.1.1 Test Procedure Outline - A Detailed Reliability Test Procedure Outline shall be submitted to and approved by the procuring activity prior to initiation of the tests. The Test Procedure must include, but is not limited to:

(a) Indicator duty cycle.

(b) Performance characteristics to be measured.

(c) Failure criteria.

* 4.4.3.1.2 Test Procedure - Three indicators shall be utilized for the reliability determination which shall be performed at Test Level III of Specification MIL-R-22973. The indicators shall be run until 2000 hours have been completed or until there is a total of seven failures. The test must be completed within four months of the date of the first contract delivery and repeated within thirty days. If a 500 hour MTBF has not been demonstrated, acceptance of the indicators will be withheld until corrective action is taken and its effectiveness is demonstrated. Indicators used for these tests may be refurbished and delivered upon satisfactory completion of the tests.

4.4.4 Rejection and Retest - The instructions for reworking, reconditioning, refurbishing or resubmitting rejected items shall be as specified in the General Provisions of the Contract or Order.

4.5 Test Conditions -

4.5.1 Atmospheric Conditions - Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and at a temperature of approximately 25°C (77°F). When tests are made with either, or both, the atmospheric pressure and temperature substantially different from the above values, proper allowance shall be made for the difference from specified conditions.

4.5.2 Level Position - Unless otherwise specified, the indicator shall be tested in its level position in which the upper mounting holes are on a horizontal line and the face is vertical.

4.5.3 Electrical Power - Unless otherwise specified, tests requiring the operation of the indicator shall be conducted utilizing a 200 ± 5 volts, 3 phase, 400 ± 20 cycles, ABC rotation, alternating current. The indicator operating under these conditions shall be deemed properly connected.

*4.5.4 Test Readings - Except where otherwise specified, the attitude indicator shall be vibrated before a test reading is taken. An electrical vibrator set at 30, 60 or 120 cps with a maximum amplitude of 0.002 inch when attached to the test item, shall be used.

4.5.5 Vibration Stand - Whenever a vibration stand is specified, it shall consist of a device which will vibrate at a desired frequency between 300 and 3000 cycles per minute and shall subject the indicator to such vibration that a point of the vibration stand shall describe, in a plane inclined 45 degrees to the horizontal plane, a circle of the diameter specified herein.

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4.5.6 Low Angle Scorsby - The low angle Scorsby shall consist of a device which will provide 5 to 7 oscillations per minute and subject the instrument to 3 degree roll, pitch and yaw movements. The 3 degree movement shall be considered 1-1/2 degrees on each side of each reference axis.

4.6 Test Methods -

4.6.1 Inspection Test - Each indicator shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.6.2 Initial Erection Test - The indicator shall be properly connected and shall be mounted in a level position. The spin axis of the gyro shall be at random position within the limits of plus or minus 85 degrees bank and plus or minus 82 degrees pitch of its operating vertical position. Power shall be applied to the indicator. The gyro shall erect to within plus or minus one degree in pitch and bank by the end of the initial fast erection period. The time delay period from the time the power is first applied to the end of the initial fast erection cycle shall not exceed one minute.

4.6.3 Final Erection Test - The gyro shall be allowed to erect one minute after the initial erection test. The gyro shall erect to within plus or minus 1/2 degree in pitch and bank. This test may be conducted with the gyro mounted on a low angle scorsby.

4.6.4 Power Warning Flag Test - The indicator shall be properly connected, except that no power shall be applied. Rated power shall then be applied to the indicator and the flag shall immediately disappear. The flag shall reappear within 15 seconds and remain in view until the gyro has been quick erected to the vertical. The flag shall immediately appear in the event of phase reversal and/or when any or all phases are broken. This test shall be repeated with the input power reduced to 185 volts, 400 cps and the flag shall operate as above.

4.6.5 Low Voltage Operation Test - The indicator shall be properly connected, except the voltage and frequency shall be reduced to 180 volts, 380 cps. The control shall be subjected to and meet the requirements specified in the following tests:

(a) Final erection (Except that the tolerance for pitch and bank indication shall be one degree and the total elapsed time shall be 3 minutes.)

(b) Power warning flag

4.6.6 Power Interruption Test - The indicator shall be properly connected and after 5 minutes of operation the power shall be removed and the indicator displaced through angles of plus or minus 60 degrees in pitch and plus or minus 55 degrees in roll and returned to level. Power shall be applied 30 seconds after removal. The pitch and bank axes shall indicate level within plus or minus 3 degrees and there shall be no recycling of the initial fast erection system. The power warning flag shall immediately retract and remain retracted.

4.6.7 Current Test - Electrical power of 200 ± 5 volts, 3 phase, 400 ± 20 cycles, phase rotation ABC, alternating current shall be applied to the AN connector of the indicator. Upon initial fast erection the current should not exceed 0.450 ampere in the A phase and 0.30 ampere in the B and C phases. After 5 minutes of operation, the current should not exceed 0.150 ampere in the A phase and 0.125 ampere in the B and C phases during normal slow erection. When the demand erection pushbutton is depressed, the current should not exceed 0.450 ampere in the A phase and 0.300 ampere in the B and C phases.

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4.6.8 Erection Rate Test - The indicator shall be mounted in a level position and operating at equilibrium. The gyro axis shall be precessed or displaced to give a roll indication slightly beyond 10 degrees in right bank. The time required for the instrument to erect from 10 to 5 degrees shall be recorded. The computed erection rate with compensation for the earth's rotation, shall be between the limits of 1.5 degrees and 2.5 degrees per minute. The same test shall be conducted for left bank, climb and dive and the same tolerance shall apply. The gyro shall not deviate in the undisplaced axis while it is erecting by more than one degree. The erection cutoff system must be by-passed for this test. This test may be conducted with the gyro mounted on a low angle scorsby.

* 4.6.9 Erection Cutoff System Test - The indicator shall be mounted in a level position and allowed to reach equilibrium. A suitable voltmeter shall be connected to the pins leading to the erection cutoff turn sensitive switch. The gyro shall be processed so that the indicator reads a left roll attitude until the voltmeter shows a steady voltage output. The angle the gyro is precessed shall be the cutoff angle of the indicator. The same procedure shall be used for right bank. The cutoff angle shall be 6 ±1 degrees.

4.6.10 Demand Erection Test - The indicator shall be mounted in a level position and operating at equilibrium. The gyro axis shall be precessed or displaced to give a roll indication slightly beyond 15 degrees in right bank. The pushbutton shall be depressed momentarily and the time required for the indicator to erect from 15 degrees to 5 degrees recorded. The computed erection rate shall be not less than 20 degrees per minute. The same test shall be conducted for left bank, climb and dive and the same tolerance shall apply. The power warning flag shall immediately appear when the pushbutton is depressed and remain in view until the gyro is erect.

4.6.11 Roll, Pitch and Yaw Test - The indicator shall be mounted in a level position on a Scorsby and operating at equilibrium. The indicator shall then be subjected to a 15 degree roll, pitch, and yaw movement for 10 minutes. At the end of this period, the indicator shall be returned to the level position, and the indicator shall indicatoe within plus or minus 1/2 degree of its zero position in roll and pitch.

* 4.6.12 Drift Test - The indicator shall be mounted on a low angle Scorsby, and shall be properly connected. The erection system shall be disconnected. With the connector end of the indicator pointed to the geographic west, the spin axis shall be displaced approximately 20 degrees from the vertical with the top of the axis in an eastward direction. Drift readings shall be recorded after a minimum period of 5 minutes of operation. With proper correction for the earth's rotation, the computed drift rate in the bank axis shall not exceed 15 degrees per hour, and in the pitch axis the computed drift rate shall not exceed 30 degrees per hour. The test shall be repeated with the indicator pointed to the south, east, and north. The same tolerance shall apply. If the drift in one heading exceeds the limit by not more than one degree in 5 minutes, the indicator may be retested in that heading and accepted if the first repeat test is in limits.

4.6.13 Loop and Roll Test - The indicator shall be mounted in a level position on a test fixture permitting rotation about the horizontal and lateral axis and allowed to reach equilibrium. The indicator shall be looped through one complete revolution (inside loop) at a rate of 10 ±1 degrees per second. The indicator shall indicate zero roll and pitch within 3 degrees immediately after this maneuver. The indicator shall then be rolled through one complete revolution at a rate of 10 to 20 degrees per second. The indicator shall indicate zero roll and pitch within one degree immediately after this maneuver.

* 4.6.14 Dielectric Strength Test - A potential of 220 volts (root mean square) alternating current at commercial frequency shall be applied between isolated pins, and between pins and the case for a period of 10 seconds. There shall be no breakdown of insulation or any other permanent damage to the instrument as a result of this test.

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4.6.15 Pitch Scale Sensitivity Test - The indicator shall be mounted in a level position on a tilt table and allowed to reach equilibrium. The displacement of the horizon line of the sphere for angles of climb shall be downward and for angles of dive shall be upward. The table shall be tilted so that the pitch scale reads 5, 10, 15, 20, 30, 40, 50, 60, 70, 80 degrees in climb and then in dive. The angles which the table has been tilted shall agree with the respective readings of the pitch scale within plus or minus 1/2 degree for pitch scale indication up to and including 20 degrees, and within plus or minus one degree for greater pitch indications.

4.6.16 Synchro Transmitting Error (Types II and IV only) Test - The indicator under test shall be properly connected and allowed to reach operating speed. A remote indicating system of the servo type, with synchro control transformer input, electronic amplification, and motor driven output, shall be properly connected to the indicator under test. The remote indicating system shall be set to read zero when receiving signals from a properly zeroed synchro transmitter. With the indicator under test in a level position the remote indicator shall indicator $0 \pm 1/2$ degree in pitch and roll. A tolerance of plus or minus one degree of error shall apply to all positions of the indicator under test up to plus or minus 80 degrees in pitch and plus or minus 180 degrees in roll.

4.6.17 Illumination Test - The indicator shall be illuminated by means of its integral lighting system and tests shall be conducted to determine conformance with Specification MIL-L-25467, except that colors shall be as noted in figure 3.

4.6.18 Magnetic Effect Test - The indicator shall be held in various positions with its nearest part 5-1/2 inches from the center of a short bar-compass whose compensator magnets have been removed, and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The various positions shall be so chosen as to include that position which causes the maximum deflection of the compass. The maximum deflection shall not exceed 5 degrees. This test shall be performed first with no voltage applied to the indicator and then shall be repeated with the indicator operating.

4.6.19 Vibration Error Test - The indicator shall be mounted on a vibration stand. The instrument shall be operated at equilibrium and the roll and pitch indications noted. The stand shall then subject the indicator to vibration, as specified herein, with a circle diameter between 0.003 and 0.005 of an inch at frequencies between 300 and 3000 cycles per minute. Throughout this frequency range the maximum total amplitude of oscillation of the bank index and the horizon line shall not exceed 2 degrees. Also, throughout this frequency range, the maximum variation of the bank index and horizon line during vibration from their original position before vibration shall not exceed 1 degree. The frequency range of 300 to 3000 cycles per minute shall be varied at a uniform rate of 90 cycles per minute per minute for a total test duration of one hour, including variations from 300 to 3000 and return to 300 cycles per minute.

4.6.20 Low Temperature Operation Test - (This test may be combined with the Low Temperature Exposure-High Altitude test at the discretion of the Testing Activity.) The indicator shall be properly connected, except that no power shall be applied. The indicator shall then be subjected to an ambient temperature of -55° ±2°C (-67°±3.6°F) for a period of 4 hours. At the end of this period, and with the temperature maintained at -55°C, power shall be applied to the indicator. The indicator shall be subjected to and meet the requirements specified in the following test:

(a) Initial erection (Except the initial erection time shall not exceed 90 seconds),

(b) Erection rate (Except the erection rate shall be not less than one degree per minute, and the deviation in the undisplaced axis shall not be greater than 3 degrees)

(c) Demand erection (Except the demand erection rate shall be not less than 15 degrees per minute)

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4.6.21 High Temperature Operation Test - (This test may be combined with the High Temperature Exposure Test at the discretion of the Testing Activity.) The indicator shall be properly connected except that no power shall be applied. The indicator shall then be subjected to an ambient temperature of 70° ±2°C (158° ±3.6°F) for a period of 4 hours. At the end of this period, and with the temperature maintained at 70°C, power shall be applied to the indicator. The indicator shall be subjected to and meet the requirements specified in the following tests:

(a) Initial erection

(b) Erection rate (Except the erection rate shall be not less than one degree per minute and the deviation in the undisplaced axis shall not be greater than 3 degrees.)

(c) Demand erection (Except the demand erection rate shall be not less than 15 degrees per minute.)

4.6.22 Seal Test - The indicator shall be tested for leaks by means of a mass spectrometer type of helium leak detector, and the differential pressure existing when the test is conducted shall be the same as that to which the instrument is filled, approximately one atmosphere. The leak rate shall be based on a one atmosphere differential pressure and pure helium as the filling medium. Under these conditions (one atmosphere differential pressure and pure helium as the filling medium), the leak rate shall not exceed 1.0 micron cubic foot per hour. Equivalent helium leak rates for the same hole size at other differential pressures or helium carbon dioxide mixtures, or both, shall be calculated by the manufacturer. The seal test shall be conducted either with the instrument evacuated and surrounded by helium or after the instrument has been filled. As an alternate test method, the indicator shall be immersed in de-aerated distilled water. The distilled water shall be treated by adding approximately 0.1 percent by weight of a wetting agent of the alkyl-aryl-sodium-sulfonate type (40 percent active ingredient). The absolute pressure of the air above the water shall then be reduced to approximately one inch of mercury. The pressure shall be maintained until air bubbles substantially cease to be given off by the water. The absolute pressure shall be increased to 2-1/2 inches of mercury and any bubbles coming from within the sealed container case shall be considered as leakage and shall be cause for rejection. Bubbles which are the result of entrapped air on the various exterior surfaces of the case shall not be considered as leakage. If the indicator incorporates attachments which are not sealed, these attachments may be removed prior to this test.

4.6.23 Radio Noise Interference Test - The electrical system of the indicator shall be tested for conducted and radiated radio noise interference in accordance with Specification MIL-I-6181.

4.6.24 Low Temperature Exposure - High Altitude Test - The indicator shall be properly connected, except that no power shall be applied. The indicator shall then be subjected to an ambient temperature of -65 ±2°C (-85° ±3.6°F) for a period of 48 hours. At the end of this period the temperature shall be raised to -55° ±2°C (-67° ±3.6°F) and maintained for an additional 24 hour period. At the end of this period, and with the temperature maintained at -55°C, the indicator shall be subjected to a simulated altitude of 80,000 ±500 feet and power shall be applied. The indicator shall be subjected to and meet the requirements of the following tests:

(a) Initial erection (Except the initial erection time shall not exceed 90 seconds)

(b) Erection rate (Except the erection rate shall be not less than one degree per minute, and the deviation in the undisplaced axis shall not be greater than 3 degrees)

(c) Demand erection (Except the demand erection rate shall be not less than 15 degrees per minute)

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The indicator shall then be returned to standard atmospheric conditions and to room temperature and allowed to saturate for a minimum period of 4 hours. The indicator shall be subjected to and meet the requirements specified in the individual tests. The indicator shall be examined and there shall be no damage which would adversely affect subsequent operation.

4.6.25 High Temperature Exposure Test - The indicator shall be properly connected, except that no power shall be applied. The indicator shall then be subjected to a temperature of 70° ±2°C (158° ±3.6°F) for a period of 24 hours. At the end of this period and with the temperature maintained at 70°C, power shall be applied. The instrument shall be subjected to and meet the requirements specified in the following tests·

(a) Initial erection

(b) Erection rate (Except the erection rate shall be not less than one degree per minute and the deviation in the undisplaced axis shall not be greater than 3 degrees.

(c) Demand erection (Except the demand erection rate shall be not less than 15 degrees per minute)

The indicator shall then be returned to room temperature and allowed to saturate for a minimum period of 4 hours. It shall then be subjected to and meet the requirements specified in the individual tests. The indicators shall be examined, and there shall be no damage which would adversely affect subsequent operation.

4.6.26 Vibration Failure Test - The indicator shall be mounted on a vibration stand and properly connected with power applied. The stand shall then subject the indicator to vibration, as specified herein, with a circle diameter between 0.018 and 0.020 of an inch. The frequency shall be varied at a uniform rate of 90 cycles per minute per minute (a variation of 2700 cycles), from 300 to 3000 cycles per minute and 3000 to 300 cycles per minute once each hour during this test. The test shall be repeated a sufficient number of times to extend the total time to approximately 3 hours (3 cycles). At the end of this period, the indicator shall be examined and there shall be no looseness in the mechanism as a result of the vibration, and there shall be no damage which would adversely affect subsequent operation. The indicator shall be subjected to and meet the requirements specified in the individual tests.

4.6.27 Humidity Test - The indicator, with no power applied but external connections made to simulate installed conditions, shall be mounted in its normal operating position and subjected to an ambient temperature of 70° ±2°C (158° ±3.6°F) and a relative humidity of 95 ±5 percent for a period of 6 hours. Distilled water shall be used to obtain the required humidity. At the end of the 6-hour period, the heat shall be shut off and the indicator shall be allowed to cool for a period of 18 hours in this atmosphere, in which the humidity rises to 100 percent as the temperature decreases. This cycle shall be repeated a sufficient number of times to extend the total time of test to 120 hours (5 cycles). Within one hour after the end of the 120-hour (5 cycles) period, the indicator shall be subjected to and meet the requirements specified in the individual tests. Examination for corrosion and other deterioration shall be made between 24 and 48 hours after completion of the humidity exposure. There shall be no corrosion or other deterioration which would adversely affect subsequent operation of the indicator.

* 4.6.28 Fungus Resistance Test - The attitude indicator shall be tested in accordance with Procedure I of Specification MIL-E-5272, except that the test period shall be 14 days. At the option of the qualifying agency, tests of component parts of the item may be accepted in lieu of, or in addition to, tests of the assembled instrument.

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* 4.6.29 Salt Spray Test - The indicator, with no power applied, but with external connections made to simulate installed conditions, shall be subject to a salt spray in accordance with Procedure I of Specification MIL-E-5272. At the end of the 50 hour period, the indicator shall be examined and there shall be no corrosion or deterioration which would adversely affect subsequent operation of the indicator. The indicator shall be subjected to and shall meet the requirements specified in the individual tests.

4.6.30 Mounting Lugs Test - The indicator case shall be mounted face downward on the removable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for one minute to each lug in a direction toward the front of the case. The lugs shall withstand the applied load without fracture.

* 4.6.31 Flight Test - Flight tests of the indicator shall be made using the turn sensitive cutoff system.

4.6.31.1 Turn Error - The indicator shall be subjected to a flight test consisting of constant speed 180 degree turns at rates of 180 and 90 degrees per minute with true airspeeds of between 100 and 500 knots. The error immediately at the completion of the turn shall not exceed one degree in pitch and/or roll. The indicator shall be subjected to additional flight tests consisting of a constant speed 180 degree turns at 90 degrees per minute with true airspeeds between 500 and 700 knots. The error immediately at the completion of the turn shall not exceed one degree in pitch and/or roll.

4.6.31.2 Loop and Roll - The indicator shall be subjected to a flight test consisting of inside loops and right and left rolls. The error in pitch and/or roll immediately after each loop, shall not exceed 10 degrees. The error in pitch and/or roll immediately after each roll shall not exceed 2 degrees and the gyro shall indicate the vertical within 1/2 degree of pitch and roll in not more than 2 minutes after the completion of the maneuver. The lateral axis of the airplane shall remain horizontal within 2 degrees throughout loops. The loops shall be made at a rate of 10 degrees per second or less.

4.6.32 Life Test - The indicator shall be properly connected and mounted on a low angle Scorsby and subjected to a cycle consisting of 2-1/2 hour period of operation and a 1/2 hour period with no power applied. During the operating period the Scorsby shall subject the indicator to roll, pitch, and yaw movements. This cycle shall be repeated a sufficient number of times to extend the operating time of the indicator to a total of 1000 hours. There shall be no failure of the indicator during the test and at the end of the 1000 hours the indicator shall be subjected to and shall meet the requirements of the individual tests.

5. PREPARATION FOR DELIVERY

5.1 Packaging -

* 5.1.1 Level A - The attitude indicator shall be packaged in accordance with Specification MIL-P-7936. The method of preservation shall be in accordance with Specification MIL-P-116, Method IA5, omitting preservation compound and using metal reusable containers (see 6.2). The metal container shall be durably and legibly marked with the following information in such a manner that the markings will not become damaged when the containers are opened:

INDICATOR: ATTITUDE, SPHERICAL, DIRECT READING,
FAST ERECTING, ELECTRIC GYRO

MS28086- (Insert proper dash number)
Date of Manufacture

* 5.1.2 Level C - Whtn this level is required, packaging shall be in accordance with standard commercial practice.

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5.2 Packing -

* 5.2.1 Levels A and B - The instrument shall be packed in accordance with Specification MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract.

* 5.2.2 Level C - The instrument shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carriers rules and regulations in effect at the time of shipment.

* 5.3 Marking - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

* 5.3.1 Precautionary Marking - The following precautionary marking shall appear on two opposite sides of each interior package and shipping container whenever practicable depending on the size of the carton:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended Use - The spherical, direct reading, fast erecting, electric gyro attitude indicators covered by this specification are intended for use in aircraft to indicate the attitude relative to the vertical during all maneuvers of the airplane in which the indicator is installed.

* 6.2 Ordering Data - Procurement documents should specify the following:

(a) Title, number and date of this Specification.

(b) The quantity, MS part number and type number of the instrument desired.

(c) Whether Sampling Plan B is to be omitted and if not, add the name and address of the laboratory to which the samples should be submitted.

(d) Levels of packaging (see 5.1) and packing (see 5.2) desired.

* 6.3 Drawing Requirements - The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-70327.

6.4 Definitions -

6.4.1 Hermetic Seal - A hermetic seal is defined as a perfectly closed and airtight seal made between metallic and metallic, or metallic and vitric materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.5 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity.

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6.6 An asterisk preceding a paragraph number denotes a change in that paragraph from the basic specification.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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